

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of this project is to automate the credit card approval process using machine learning techniques, enabling faster, more accurate, and data-driven credit approval decisions.
Scope	The project focuses on analyzing applicant financial and demographic information to predict whether a credit card application should be approved or rejected. The system integrates machine learning models with a Flask web application and supports cloud deployment for real-time predictions.
Problem Statement	
Description	Traditional credit card approval processes rely heavily on manual verification and predefined rules, which are time-consuming, prone to human error, and inefficient when handling large volumes of applications.

Impact	Addressing these challenges will improve operational efficiency, reduce processing time, minimize approval errors, enhance risk assessment, and provide a better customer experience for both applicants and financial institutions.
Proposed Solution	
Approach	The proposed solution uses machine learning algorithms to analyze applicant information such as income, employment status, education level, credit history, and financial records to predict credit card approval eligibility.
Key Features	• Implementation of a machine learning-based credit card approval prediction model. • Real-time approval/rejection prediction through a Flask web application. • Comparison of multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, and XGBoost



	• Selection and deployment of the best-performing model. • Cloud deployment using IBM Watson Machine Learning. • User-friendly interface for applicants and bank analysts. • Prediction result storage for future analysis and auditing.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources	Intel i5/i7 Processor or equivalent
	Memory	8 GB RAM

	Storage	256 GB SSD or higher
Software	Frameworks	Flask
	Libraries	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
	Development Environment	Jupyter Notebook, VS Code, PyCharm
Data	Dataset Source	Kaggle Credit Card Approval Dataset
	Dataset Format	CSV
	Data Attributes	Gender, Income Type, Education Level, Employment Duration, Credit History, Loan Records, Family Status, Annual Income